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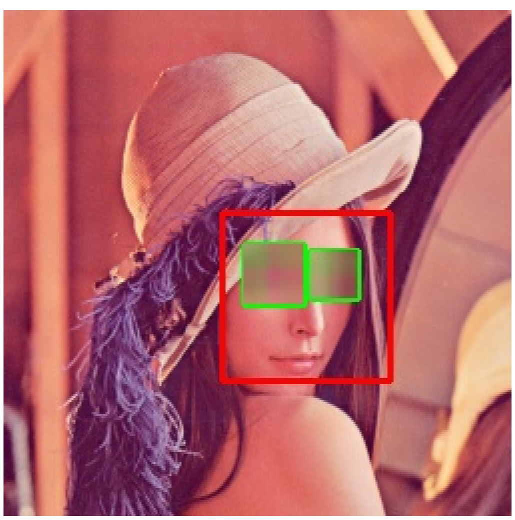
599 Edge AI and Robotics
Module 3 Face and Eye Recognition Exercise

Overview:
This is a brief exercise that is just an extension of the previous lab. The difference being instead of redacting the entire face, only the eyes are redacted.

Implementation Steps:

- Read the image using the cv2.imread()
- Convert the image from BGR to RGB for cv2
- Convert the image to grayscale
- Get the filepaths for the xml files containing the haar cascade data
- Create a cv2.CascadeClassifier with the path for the face and eye xml data
- Run the detection to identify the face
- For each face found, draw the bounding box with cv2.rectangle and identify the eyes
- For each eye found, use cv2.rectangle to draw the bounding box
- Fill the found region using the cv2.GuassianBlur
- Convert image back to BGR
- Display image as ipywidget

Results:



The exercise also contains a GPU implementation, though similar to the last lab the CPU still draws the bounding boxes and applies the blur, the GPU just detects the face. This is as simple as enabling the CUDA classifier and passing the image to/from the device.

GPU result: